

► **COVID-19 DEMONSTRATED** the world's dependence on semiconductors. Even as major chip fabrication units, or fabs, continued operating as essential infrastructure, disruptions across logistics, materials, assembly and testing, combined with demand swings, took a toll on availability. The shortage rippled through sectors ranging from consumer electronics and automobiles to industrial equipment and national defence, laying bare the chinks in global supply chains.

Indian industry, too, felt the impact. It was against this backdrop that Arun Ramchandani, Head of L&T Precision Engineering and Systems, called his old classmate and long-time friend Sandeep Kumar, a 42-year veteran of the semiconductor industry. Ramchandani posed a simple question: What could they do for India?

What followed was nearly two years of discussions, refinements and multiple iterations of plans presented to L&T's leadership. Finance and marketing teams worked on their own strategies in parallel. By late 2023, a clear direction had emerged.

L&T chose to tread a different path from chip fabs or outsourced semiconductor assembly and testing (OSAT). In November 2023, it founded L&T Semiconductor Technologies Ltd (LTSCT). The mandate was a laser focus on designing chips. "Product is where the IP (intellectual property) is, product is where the differentiation is," says Kumar, Chief Executive of LTSCT. "Once we sell enough products, we will build factories to make those products."

L&T is not alone. Encouraged by India's semiconductor push that emphasises building chip fabs and backend manufacturing capacity, Indian conglomerates are shifting from design and engineering services to silicon ownership. But L&T's approach highlights a quieter reality. Fabs can cost \$10–20 billion and typically operate on thin margins but design-led companies capture the bulk of the economic value. The world's top 10 semiconductor firms have a combined market capitalisation of roughly \$10.3 trillion, says US based semiconductor consultancy

Fab Economics. More than half of them derive their core value from chip design, even when manufacturing forms part of their operating model for some. Like L&T, other Indian industrial groups are tweaking their plans to de-risk and capture a bigger part of the value chain.

ENABLERS TO OWNERS

The Murugappa Group initially entered the semiconductor ecosystem through testing and packaging with CG Semi. But the group soon recognised that backend capabilities alone will not deliver strategic control or long-term differentiation. The next step was chip design. This led to the creation of Axiro Semiconductor Private Ltd.

"Manufacturing is essential, but it is design and ownership of the product that determines architecture, performance, differentiation and long-term value," says Naveen Yanduru, CEO of Axiro Semiconductor. "Over the last decade, especially post-pandemic, industries have realised that resilience is not only about fabs, it is also about owning critical design capabilities and understanding how chips integrate into real systems."

India's new focus on chip design builds on decades of experience in the field. Over the years, it has developed deep expertise at the foundational layer of the value chain. The country accounts for nearly one-fifth of the world's chip design talent, which contributes to some of the most advanced microchips being produced across the world.

In February, Qualcomm of the US announced the successful tape-out of a two-nanometer (nm) semiconductor design, with the effort led in large part by its engineering teams in Bengaluru, Chennai and Hyderabad. Tape-out marks the final stage of chip design, when design data is transferred to a foundry such as Taiwan Semiconductor Manufacturing Company (TSMC) for manufacturing.

Some Indian IT companies have been providing semiconductor engineering services to global clients



WITH ISM 2.0, THE GOVERNMENT'S FOCUS IS SHIFTING TOWARDS CHIP DESIGN AND IP CREATION. THIS MIGHT OPEN THE DOOR FOR LARGE PLAYERS